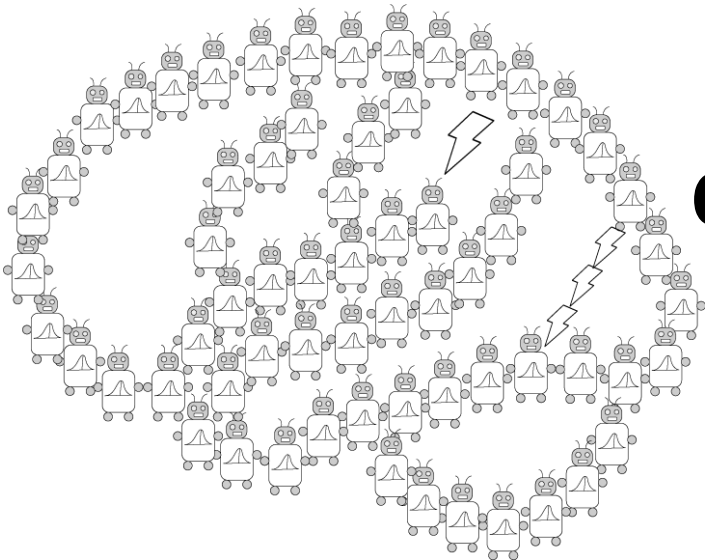


Basic Bayes and Judgment



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

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January 23, 2025

- Foundations in probability
 - Definitions: Set, outcome set/space, size, event, event set/space, random variables, and probability.
 - More on probability
 - Joint probability
 - Conditional probability as set restriction
 - Traditional definition of conditional probability
 - Marginalization and proving Bayes rule
- Intuitive Bayes
- Human judgment and heuristics and biases
- Conclusions



Foundations of Probability

A **set** is an **unordered collection of items** and its **elements are unique**.

So, $\{\{HH, HT\}, \{HH, HT\}, \{TT\}\} = \{\{HH, HT\}, \{TT\}\}$

Outcomes are either **in** or **out** of a set.

$$HH \in \{HH, HT, TH\} \quad TT \notin \{HH, HH, TH\}$$

“in” operator

“not in” operator



Foundations of Probability

Outcome set/space: Set/space of possible outcomes for all variables

E.g., For an outcome set of two coin flips:

$$S = \{HH, HT, TH, TT\}$$

Start and
end a set

$|\cdot|$ is the **cardinality** or **size** of a set.

E.g., $|S| = 4$ because there are 4 elements.

Event: a set of outcomes.

E.g., “at least one head” = $\{HH, HT, TH\}$

Event space: Set/space of events

E.g., $E = \{ \{HH, HT\}, \{HH, HT, TH\}, \{HH\}, \{\} \}$

Empty or
null set
(set with no
items)



Foundations of Probability

Event set/space: Set/space of events

E.g., $E = \{ \{HH, HT\}, \{HH, HT, TH\}, \{HH\}, \{\} \}$

For finite outcome sets, all possible events is often the event space.

$$E = \{ \{\}, \{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \\ \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}, \\ \{HH, HT, TH\}, \{HH, HT, TT\}, \{HH, TH, TT\}, \\ \{HT, TH, TT\}, \{HH, HT, TH, TT\} \}$$

For continuous outcomes, the word “space” is typically used instead of set. The math gets trickier.

We will discuss this more next week.



Foundations of Probability

A **random variable** is a **function** from the **sample space** to (an) **event space**.

E.g., Let Y be at least one of two coinflips is Heads.

It takes an outcome (e.g., HT) and puts it in its event space. What is the event space here?

$$\{\text{True}, \text{False}\} = \{T, F\}$$

Definition: A **probability** counts the number of outcomes that fall into one value as opposed to the number of possible outcomes.

What is the probability that $Y = \text{True}$, $P(Y = \text{True})$, or $P(Y = T)$ or $P(Y)$?

HH	HT
TH	TT



Foundations of Probability

First definition: A **probability** counts the number of outcomes that fall into one value as opposed to the number of possible outcomes.

Let C_n be the outcome of the n -th coin.

So, C_1 is the outcome of the 1st coin.

What is the probability that the first coin is heads?

$$P(C_1 = H) =$$



Foundations of Probability

Joint probability: What is the probability of two (or more) random variables have particular values (or satisfy some condition) for the same outcome?

E.g., what is the probability that the first coin is heads and the second coin is tails?

$$P(C_1 = H, C_2 = T) = P((C_1 = H) \cap (C_2 = T))$$

“Intersects” or “AND”
(what’s in both sets)

$$\{HH, HT\} \cap \{HT, TT\} = \{HT\}$$

$$P(C_1 = H, C_2 = T) =$$



Foundations of Probability

Conditional probability: What is the probability of one or more variables having some values given one or more other variables have some other values?

It is the probability of the first random variable(s) when the universe of discourse is the set of outcomes that satisfy the second random variable(s).

E.g., what is the probability of the second coin being tails **given** the first coin was heads?

$$P(C_2 = T | C_1 = H)$$

1/2

HH	HT
TH	TT



More on Conditional Probability

Outcome set/space: Set/space of possible outcomes for all variables (worlds)

E.g., Outcome set for two coin flips:

$$S = \{HH, HT, TH, TT\}$$

Remember our variables:

Let C_n be the outcome of the n -th coin.

So, C_1 is the outcome of the 1st coin.

E.g., X_1 is "at least one head".

What is the probability that the first coin is heads given there was at least one heads?

$$P(C_1 | X_1 = T) = \begin{array}{|cc|} \hline HH & HT \\ \hline TH & TT \\ \hline \end{array} \quad 2/3$$



Conditional Probability Redux

What is the probability that the first coin is heads given there was at least one heads?

$$P(C_1|X_1 = T) = \begin{array}{|c|c|} \hline \text{HH} & \text{HT} \\ \hline \text{TH} & \text{TT} \\ \hline \end{array} \quad 2/3$$

Boring definition of conditional probability*:

$$P(C_1 = H|X_1 = T) = \frac{P(C_1=H, X_1=True)}{P(X_1=True)}$$

$$\begin{array}{|c|c|} \hline \text{HH} & \text{HT} \\ \hline \text{TH} & \text{TT} \\ \hline \end{array} \quad \begin{aligned} &P(X_1 = T) \\ &(C = H) \cap (X_1 = T) = \\ &\Rightarrow P(C_1 = H, X_1 = True) = \end{aligned}$$

3/4



Marginalization

If you have two random variables, X_1 and C_1 , and you are interested in the probability of just X_1 having a value (True), you can calculate it by adding over the joint probability of X_1 having that value and each value of C_1 .

Sum

Values summed over

$$P(X_1 = T) = \sum_{a \in \{T, F\}} P(X_1 = T, C_1 = a)$$

$$P(X_1 = T) = P(X_1 = T, C_1 = T) + P(X_1 = T, C_1 = H)$$

HH	HT
TH	TT

HH	HT
TH	TT



Proving Bayes' Theorem

Recall one definition of conditional probability:

For two random variables A and B, the cond. prob. of A given B is

Product rule

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

$$P(A = a|B = b)$$

Thinking
generatively...

The cond. prob. of B given A is

$$\begin{aligned} P(B|A) &= \frac{P(B,A)}{P(A)} = \\ &= P(A, B) = \end{aligned}$$

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In a dark and depressing world...

You have a 50 year-old friend who has been a chain smoker since they were 18.

There's a nasty cold going around.

You go over to their house to bring them soup.

You hear them cough.

You know that they're sick, but you don't know what it is.

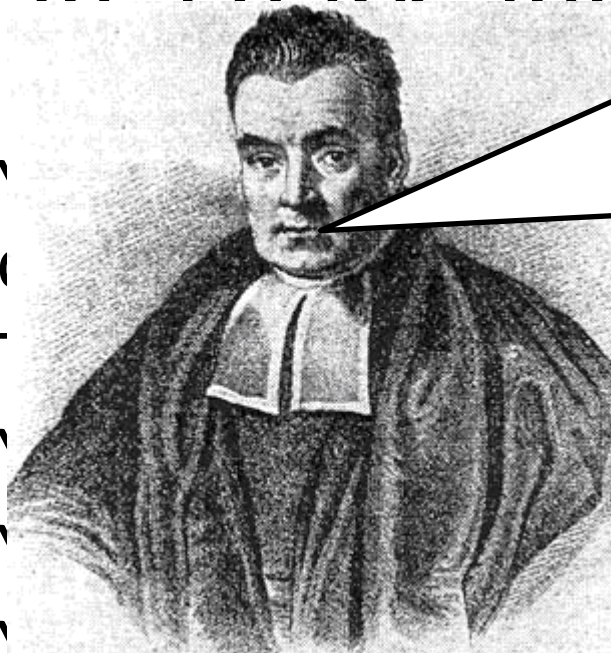
You've limited it to three possibilities:

A cold, a stomach virus, or lung cancer

How likely is each?



In a dark and de



Reverend Thomas Bayes
(1701-1761)

I've got a rule to help!

I used it to try to prove G-d exists. It
made me feel better.

they

d goin

r hous

gh.

're sick,

Turns out that it's really useful for other
stuff too, but I didn't figure that out!

For you to use it, you'll need to learn to
write probabilities carefully.

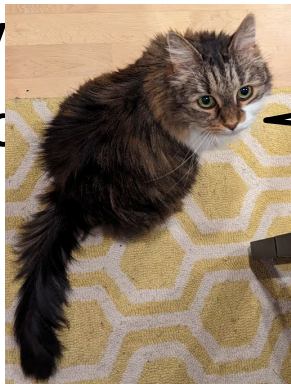
Math is so cool! Try this at parties.

you don't know

You've

A c

How



it to thro

made

ach?

I'm unimpressed.

Please feed me treats.



Writing probabilities

d is a hidden property you don't get to observe.

E.g., disease, emotions, goals...

Writing probabilities

d is a hidden property you don't get to observe.

E.g., disease, emotions, goals...

o is what you observe in the world

E.g., coughing, punching, grabbing milk, ...

$P(d = \text{lung cancer})$ means the probability of lung cancer

“prior probability”

$P(d = \text{lung cancer} \mid o = \text{cough})$ means the probability of lung cancer given you know (things to right of “ $\mid$ ”) that the person coughed.

“posterior probability”

Writing probabilities

d is a hidden property you don't get to observe.

E.g., disease, emotions, goals...

o is what you observe in the world

E.g., coughing, punching, grabbing milk, ...

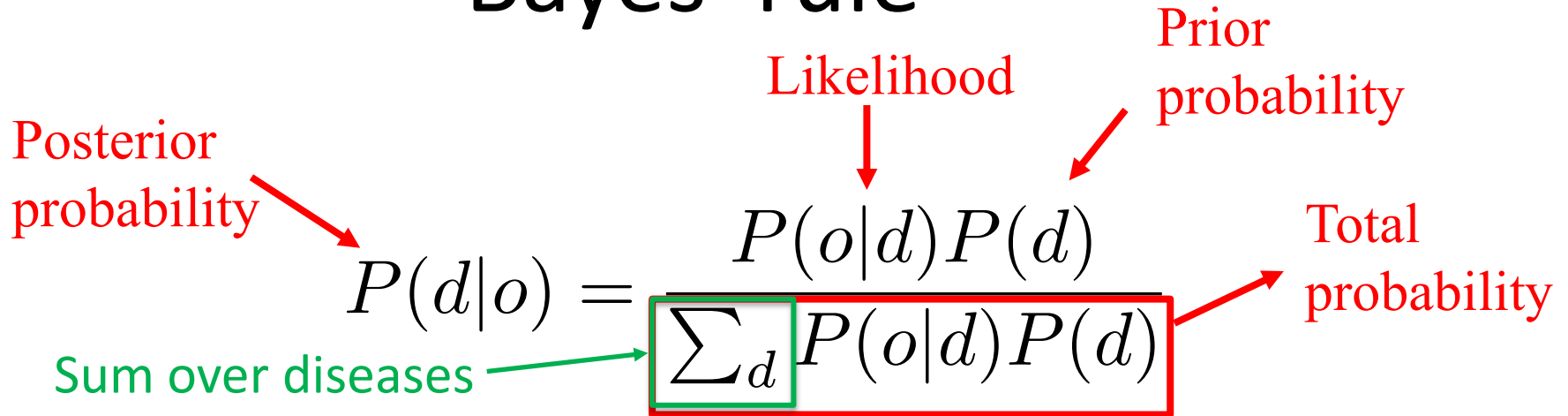
$P(d = \text{lung cancer} \mid o = \text{cough})$ means the probability of lung cancer given you know (things to right of “ $\mid$ ”) that the person coughed.

“posterior probability”

$P(o = \text{cough} \mid d = \text{lung cancer})$ means the probability of coughing given you know (things to right of “ $\mid$ ”) that the person has lung cancer.

“likelihood”

Bayes' rule



The diagram illustrates Bayes' rule with the following components and annotations:

- Posterior probability:** $P(d|o)$
- Likelihood:** $P(o|d)$
- Prior probability:** $P(d)$
- Sum over diseases:** $\sum_d$
- Total probability:** $P(o|d)P(d)$

$$P(d|o) = \frac{P(o|d)P(d)}{\sum_d P(o|d)P(d)}$$

Given you hear your friend cough, what is the probability that she has...

$P(d|o = \text{Cough})$ for

$d_1 = \text{Cold}$

$d_2 = \text{Stomach Virus}$

$d_3 = \text{Lung Cancer}$



Bayes' rule via word math

	<u>likelihood</u> $P(o d)$	<u>prior</u> $P(d)$	<u>posterior</u> $P(d o = \text{Cough})$
$d_1 = \text{Cold}$	high	medium	high
$d_2 = \text{Stomach Virus}$	low	medium	low
$d_3 = \text{Lung Cancer}$	high	very low	low/medium



To get actual probability, you divide each of these by their sum (*normalization*).

Bayes' rule

	<u>likelihood</u> $P(o d)$	<u>prior</u> $P(d)$	<u>posterior</u> $P(d o = \text{Cough})$
$d_1 = \text{Cold}$	0.9	0.45	
$d_2 = \text{Stomach Virus}$	0.1	0.45	
$d_3 = \text{Lung Cancer}$	0.9	0.1	

To get actual probability, you divide each of these by their sum (*normalization*).

Bayes' rule

	<u>likelihood</u> $P(o d)$	<u>prior</u> $P(d)$	<u>posterior</u> $P(d o = \text{Cough})$
$d_1 = \text{Cold}$	0.9	0.45	0.405
$d_2 = \text{Stomach Virus}$	0.1	0.45	0.045
$d_3 = \text{Lung Cancer}$	0.9	0.1	0.090

To get actual posterior, you divide each of these by their sum (*normalization*).

$$0.405 + 0.045 + 0.090 = 0.54$$

Bayes' rule

	<u>likelihood</u> $P(o d)$	<u>prior</u> $P(d)$	<u>posterior</u> $P(d o = \text{Cough})$
$d_1 = \text{Cold}$	0.9	0.45	$0.405/0.54 = 0.75$
$d_2 = \text{Stomach Virus}$	0.1	0.45	0.045
$d_3 = \text{Lung Cancer}$	0.9	0.1	0.090

To get actual posterior, you divide each of these by their sum (*normalization*).

$$0.405 + 0.045 + 0.090 = 0.54$$

Bayes' rule

	<u>likelihood</u> $P(o d)$	<u>prior</u> $P(d)$	<u>posterior</u> $P(d o = \text{Cough})$
$d_1 = \text{Cold}$	0.9	0.45	$0.405/0.54 = 0.75$
$d_2 = \text{Stomach Virus}$	0.1	0.45	$0.045/0.54 = 0.083$
$d_3 = \text{Lung Cancer}$	0.9	0.1	$0.090/0.54 = 0.167$

To get actual posterior, you divide each of these by their sum (*normalization*).

$$0.405 + 0.045 + 0.090 = 0.54$$



Bayes' rule

$$\overset{\text{Posterior}}{P(h|d)} \propto \overset{\text{likelihood}}{P(d|h)} \overset{\text{prior}}{P(h)}$$

h can be... the

- Height of an object

- Sound produced by a speaker

- Time for an event to complete

- Graduate student specialization

- Engineer/Lawyer

- GPA of student

Why is this controversial as a framework for studying the mind?

- Foundations in probability
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What is reasoning and why do it?

What is reasoning?

Inferring from the known to the unknown. Called an **inference**
“Going beyond the information given”

What is the goal of reasoning?

To acquire information that supports adaptive behavior

Why is this hard?

Too much info.: problem of deciding which information is relevant.

E.g., There are small objects at the bottom and top of a hill.

What will happen? Does it matter what the objects are? Their states? The time of day? Whether there are other objects?...

Too little info.: **inferences** are often **underdetermined**
(more than one solution)...

E.g., state of the world given retinal image (visual illusions)...

Deduction and Induction

If the Prof. is great, then I am enjoying class.

The Prof. is great

I am enjoying class.

Deduction

Object 1 is a Swan and it is White

Object 2 is a Swan and it is White

Object 3 is a Swan and it is White

If X is a Swan, then it is White

Induction

Deduction = conclusions **guaranteed** to be true

Induction = conclusions **likely** to be true

Deduction vs. Induction

Deduction

- Preserves truth
- Depends on form, not on content
- Logic

Induction

- Confers probability
- Depends on form and on content
- Probability theory



“Normative theories”

Specify how one *should* make inferences

“Descriptive theories”

Describe how people *in fact* make inferences

Judgment

Judgment is the subjective/intuitive evaluation of an **event**. Often its probability (this is **inductive reasoning**)

E.g. How likely is it that:

“the defendant did it”,

“I will win if I gamble”,

“I will get cancer if I smoke”,

“Aubrey Plaza will go out with

Why is this important?

We should make decisions based on likelihood of the decision and the value of those events

It's Thursday at 6pm. Should you study tomorrow or go out with your friends





Two cab companies operate in a given city, the Blue and the Green (according to the color of cab they run).

85% of the cabs in the city are Green,
and the remaining 15% are Blue.

A cab was involved in a hit-and-run at night.

There was a witness.

The witness is able to correctly identify the color of a cab at night 80% of the time.

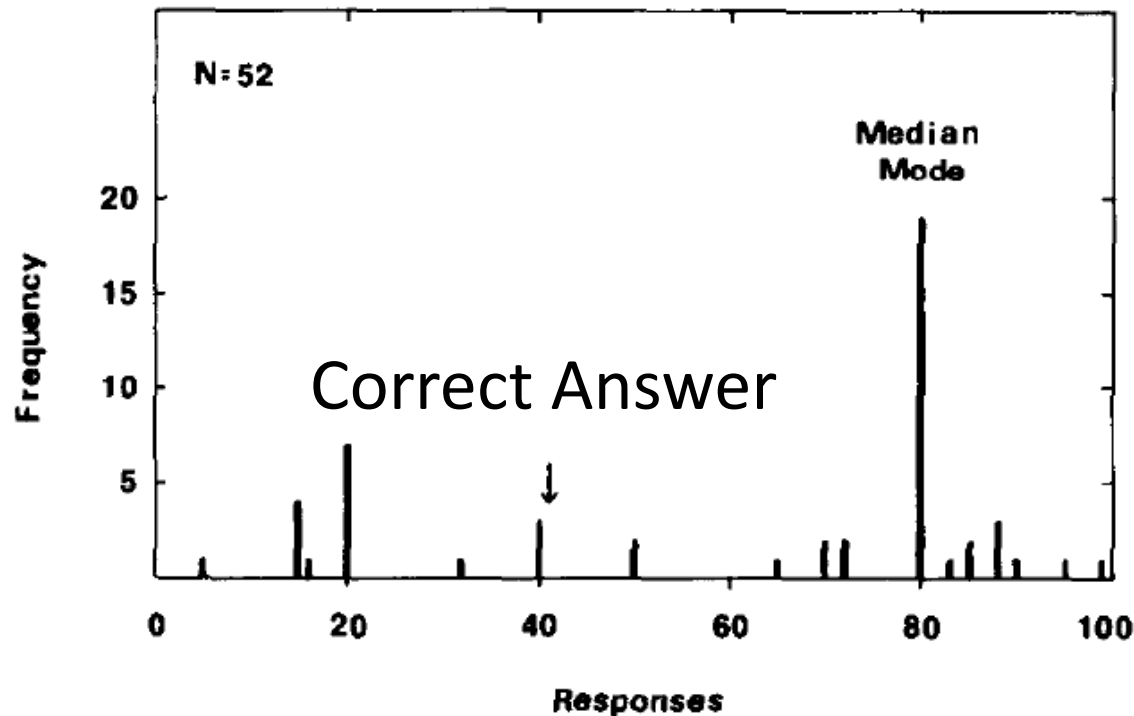
A witness identifies the cab as a Blue cab.

What is the probability that the cab is Blue?

Base-rate neglect

Participants report the **likelihood** and neglect integrating the **prior**.

Remember that they should be multiplied together... (0.8×0.15 is numerator)



Medical doctors/residents also fail at this, even in the context of medical diagnosis (Casscells, Schoenberger, & Grayboys, 1978).

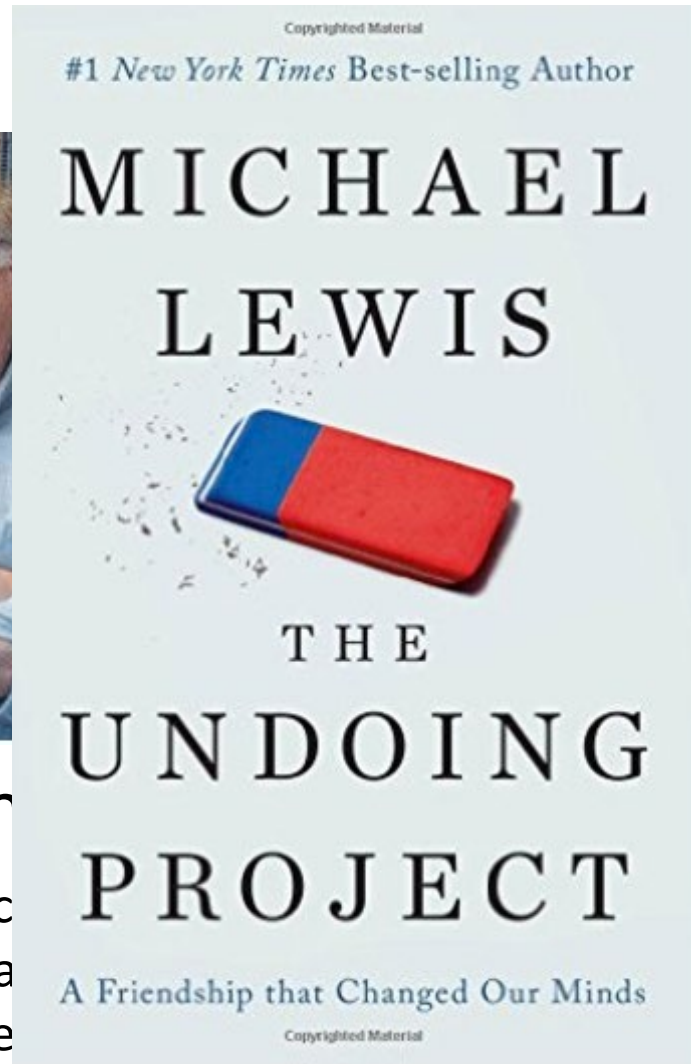
(Kahneman & Tversky, 1972; Bar-Hillel, 1980)

Heuristics and biases



Daniel Kahneman

Won Nobel prize in economics despite never taking a course in economics. Ranked 101st greatest Israeli ever.



Amos Tversky

Won MacArthur Genius Fellowship and several other awards.

Also won Nobel prize too if he hadn't passed away in 1990s.

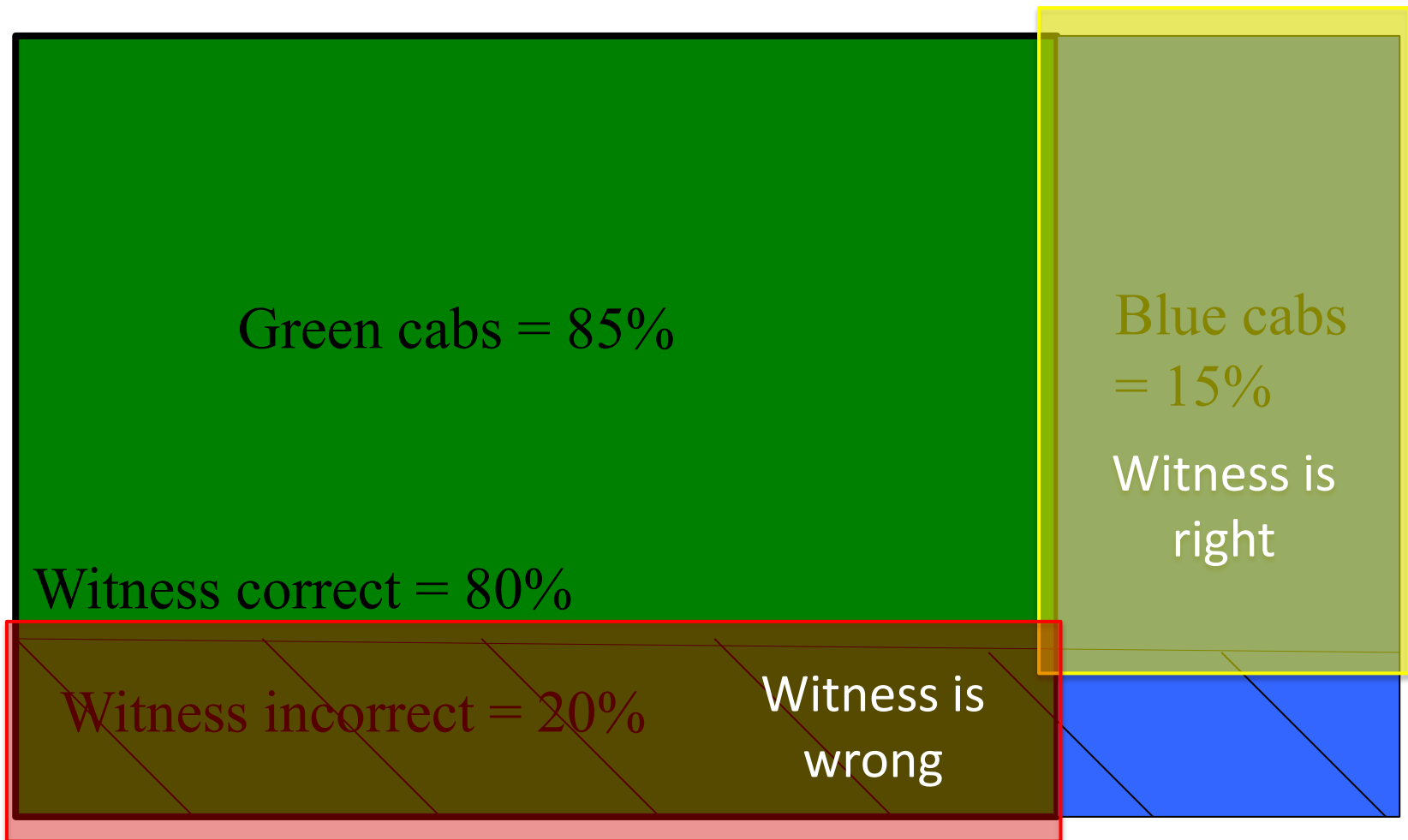
Heuristics and Biases for Cognition

- Our cognition (incl. maps) tend to be accurate
- Errors usually reflect using reasonable strategies that are inappropriate in situations that are not ecologically valid.
 - **Bias** is knowledge that produces **errors** because it is used by the mind when it may or may not be appropriate or applicable.
- A **heuristic** a rule of thumb or general problem-solving strategy that typically provides good solutions, but may contain biases.

Getting the correct answer:

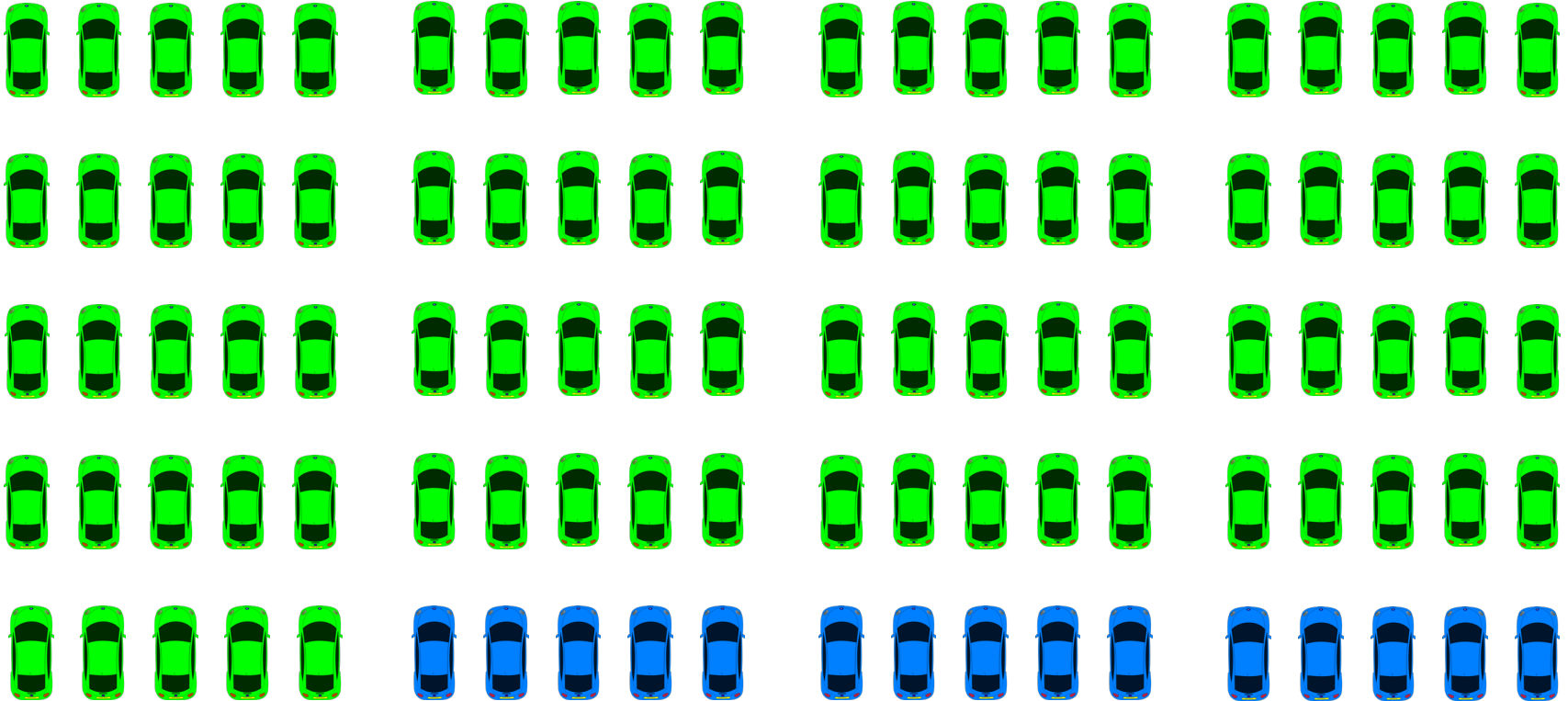
Depicting the situation (Witness said blue)

Answer is area of yellow relative to area of red.



Getting the correct answer: Another depiction

85 of 100 Green



15 of 100 Blue

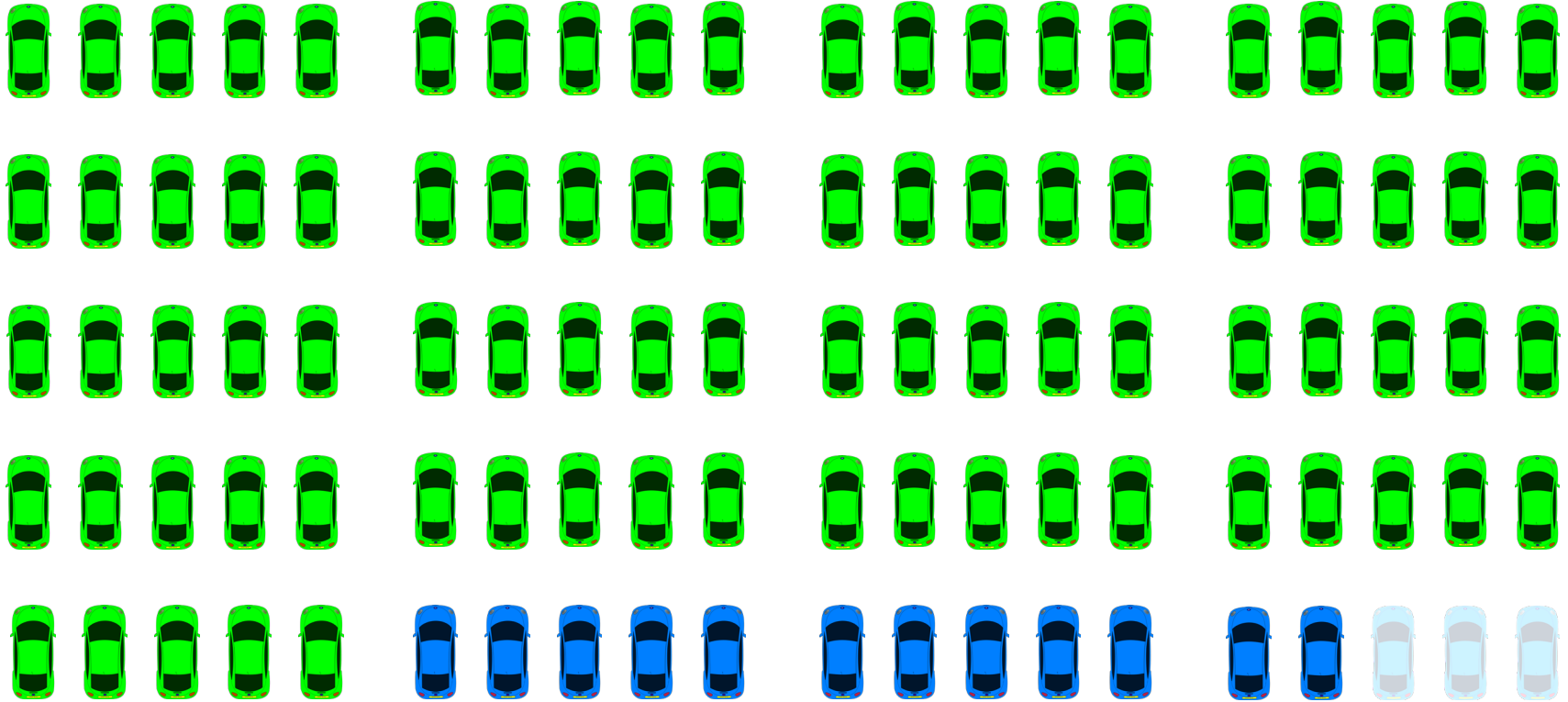
80% of 15 is 12

All the cabs in the city

Getting the correct answer: Another depiction

20% of 85 is 17

85 of 100 Green



15 of 100 Blue

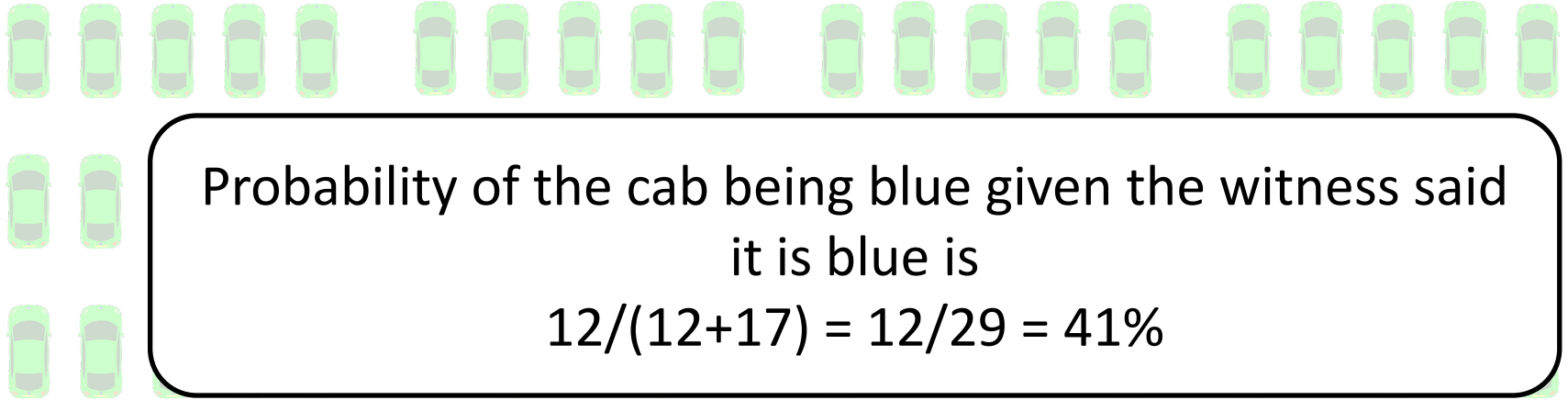
80% of 15 is 12

All the cabs in the city

Getting the correct answer: Another depiction

20% of 85 is 17

85 of 100 Green



15 of 100 Blue

80% of 15 is 12

All the cabs in the city

Bayes' rule... again

$$P(d|o) = \frac{P(o|d)P(d)}{\sum_d P(o|d)P(d)}$$

Step-by-step:

1. Identify the “*disease*” and “*observation*” for the problem.
2. Use information given to define *likelihoods* ($P(o|d)$ for each d) and *priors* ($P(d)$ for each d).
3. Calculate numerators for each d .
4. Calculate the denominator for all diseases by summing the numerators (one disease must have produced the observation).
5. Divide the numerators by the denominator to be probabilities.



Bayes' rule... again!

Observation: Witness reports blue ($W = b$) (correct 80%)

“Diseases”:

B (the cab in the accident was blue) 15% Prior

G (the cab in the accident was green) 85% Prior

$$P(B|W = b) \propto P(W = b|B)P(B) = 0.8 \times 0.15 = 0.12$$

$$P(G|W = b) \propto P(W = b|G)P(G) = 0.2 \times 0.85 = 0.17$$

⇒ Slightly more likely to be Green despite witness saying blue!

⇒ Normalize: denominator = $0.12 + 0.17 = 0.29$

$$P(B|W = b) = 0.12/0.29 = 0.41$$

$$P(G|W = b) = 0.17/0.29 = 0.59$$

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Bayesianism

Bayesian inference is the mathematics of inductive problems.*

Steps to forming a Bayesian model:

1. Formalize the problem that people face.
2. Formalize the types of knowledge people have when solving the problem.
3. Apply Bayes' rule and see what an ideal learner/observer would learn from the data.
4. How does the ideal learner behave?
5. What types of knowledge (*and/or constraints on learning) must be assumed for the model and people to behave similarly?